

Learning with less: label-efficient land cover classification at very high spatial resolution using semantic stratification and self-supervised deep learning

Author order?

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Abstract

Deep learning methods for land cover classification have been shown to be highly performant, particularly at high spatial resolutions where traditional pixel-wise or object-based approaches struggle. However, the wide parameter space of deep neural networks typically requires several thousand densely annotated imagery tiles to achieve reliable performance. This is problematic, as data annotation is costly and time-consuming, creating a strategic barrier to widespread adoption of such models for land cover classification at high resolutions. We address this challenge through two complementary approaches. First, we demonstrate that utilizing a stratification approach that groups candidate imagery tiles into semantic clusters using existing moderate resolution land cover products improves the quality of a training dataset compared to simple random sampling. Furthermore, applying Bootstrap Your Own Latent (BYOL) self-supervised learning to an ImageNet pre-trained ResNet-101 encoder using unannotated aerial imagery enhances the performance of a semantic segmentation network on a downstream land cover classification task, particularly when few samples are supplied during training. When applied to a 1 m resolution land cover task using linear probing, our BYOL model demonstrates average increases of 13.04% and 14.87% in mIoU and overall accuracy over a similar model pre-trained on ImageNet alone, illustrating the extent of the domain gap between general-purpose imagery often used for pre-training and high resolution remote sensing data. Using only 500 labeled samples during the training phase, our approach achieves 87.42% overall accuracy and 63.00% mIoU for comprehensive 8-class land cover mapping across Mississippi when the model is fine-tuned in a U-Net framework—improvements of 2.06% and 2.77% over ImageNet pre-training. These results demonstrate that creative approaches to sample stratification and selection along with advanced pre-training can substantially reduce the need for hand-annotated data, breaking down hurdles for large-scale high spatial resolution land cover mapping.

1 Introduction

Land cover data provides valuable information about the characteristics of the Earth’s surface for a wide variety of applications across many domains, including agriculture, urban planning, and environmental monitoring Ban et al. (2015); Loveland et al. (2000); Scarpace and Quirk (1980); Townshend (1992); Wulder et al. (2018). Research into automated methods for land cover

classification from remotely sensed data has been ongoing for several decades, with a majority of methods relying on supervised learning techniques to classify land cover using multispectral aerial or satellite imagery Maxwell et al. (2018); Phiri and Morgenroth (2017). Historically, pixel-wise classification methods have been the predominant approach for land cover classification, where each pixel in the image is classified based on the pixel's raw spectral values, hand-crafted features, and ancillary data from other geospatial datasets Khatami et al. (2016); McIver and Friedl (2001). This pixel-wise paradigm is effective for classifying coarse and moderate spatial resolution imagery (e.g., MODIS, Landsat, Sentinel-2), but is less effective for classifying high spatial resolution imagery due to decreased inter-class spectral separability and increased intra-class spectral variability Grekousis et al. (2015); Hansen and Loveland (2012); Lu and Weng (2007); Markham and Townshend (1981); Strahler et al. (1986); Woodcock and Strahler (1987). This is problematic, as ideally, high spatial resolution (< 10 m) land cover data offers more detailed information about the Earth's surface with greater spatial accuracy, which is valuable for applications that require fine-grained information about land cover types and their spatial distribution Fisher et al. (2018); Li et al. (2022a). Object-based Image Analysis (OBIA) techniques have been proposed as a potential workflow to address these issues, but their performance is highly dependent on the quality of objects or superpixels extracted from the imagery during the segmentation process, which must be carefully tuned to the specific characteristics of the imagery and the land cover classes of interest Blaschke et al. (2014); Johnson and Xie (2011); Ma et al. (2017). Still, a key idea behind OBIA is that the spatial information contained in the objects can be leveraged to provide more information to the classifier, which addresses the inter-class spectral separability issues that hamper pixel-wise classification methods Blaschke et al. (2000); Myint et al. (2011); Tehrany et al. (2014). The application of deep computer vision models to land cover classification has been of significant interest in recent years, as these models excel at extracting spatial and spectral features from imagery in a data-driven manner, making them well-suited for high spatial resolution land cover classification tasks Ma et al. (2019); Vali et al. (2020); Zhu et al. (2017).

Convolutional neural networks (CNNs) are a class of deep learning models that have been widely used for land cover classification tasks due to their ability to automatically extract hierarchical spatial structures from the input imagery that are useful for classifying land cover type Kussul et al. (2017); Li et al. (2022b); Pan et al. (2020a); Song et al. (2019). CNN-based semantic segmentation models learn a mapping from the input image to a pixel-wise classification probability map, which can be used to generate a land cover map by assigning each pixel to the class with the highest probability Long et al. (2015). Semantic segmentation models have been widely used for land cover classification tasks, as they can facilitate the generation of high spatial resolution land cover maps that leverage both spectral and spatial information in the input imagery under a unified framework Yuan et al. (2021). However, these models require large amounts of labeled training data to learn the complex spatial and spectral patterns present in the imagery, which can be difficult to obtain for high spatial resolution imagery due to the high cost of collecting and labeling the data Hu et al. (2015).

This need for large amounts of ground truth data for accurate training of deep learning models has led to the extensive research into methods for reducing the amount of data necessary to train these models. Most of these methods focus on leveraging transfer learning, where the parameters of a model pre-trained using a large dataset are transferred to a model that is subsequently fine-tuned on a smaller dataset of interest. Transfer learning is effective for deep computer vision models, as the features learned by a model on a large dataset are often generalizable to other datasets thanks to the

hierarchical nature of the features learned by the model Hu et al. (2015); Kolesnikov et al. (2020). However, the effectiveness of transfer learning is highly dependent on the similarity between the source and target datasets, as well as the amount of labeled data available for the target dataset. For example, a common approach to transfer learning for land cover classification tasks is to pre-train a model on a large general-purpose image classification dataset (e.g., ImageNet Russakovsky et al. (2015)) and then fine-tune the model on a smaller dataset of interest Pires de Lima and Marfurt (2020). This is generally effective, but the domain shift between the source and target datasets can hinder the performance of the model, and the extent of this domain shift between general purpose image classification datasets and high spatial resolution remote sensing datasets is not well understood Ma et al. (2024). Self-supervised pre-training methods have been proposed as an alternative to traditional fully supervised pre-training methods, as they can leverage the spatial and spectral information present in the input imagery to learn useful features through the use of pretext tasks that do not require labeled data Chen et al. (2020); Tian et al. (2020); Wu et al. (2018). In remote sensing, these methods are becoming more widely explored as a means to pre-train large foundation models that can serve as strong feature extraction backbones for downstream tasks **CITE!**. While effective, foundation models are geared for performance in multi-sensor, multi-modal, or cross-task scenarios, and not strictly label efficiency **CITE!**. As such, to the authors' knowledge, little work has been done on how these self-supervised pre-training approaches can impact operational land cover mapping under extreme low-data scenarios.

Beyond the choice of pre-training routine, another poorly understood component of developing semantic segmentation models for land cover classification is the choice of sampling routine used to select training data. Ideally, a simple random sample of sufficient size from the region of interest should produce a quality training dataset. However, the sample size must be large enough to satisfy the conditions that the sampled data is representative of the variation in land cover across the region while also having enough samples in the minority classes such that a classifier can capture said classes accurately. Simple random sampling becomes problematic for large areas with diverse land cover types or strongly imbalanced land cover distributions. Much research on stratification strategies for building pixel-based datasets for land cover classification has been conducted to address the issues associated with simple random sampling. These stratification procedures involve splitting the population into several groups based on some intrinsic property (e.g., geospatial location) or information from an external data source (e.g., prior land cover information) and sampling from these groups proportionally. However, semantic segmentation models require small patches consisting of dense labels, that is, every pixel in the patch is assigned to a class, rather than individual pixels. While this does not negate the implications of stratification, it does add extra considerations to the process, as one patch can have multiple classes. For example, when stratifying training data by land cover class according to some prior knowledge of the land cover distribution, it can be very difficult to select patches in such a way that fit the desired land cover distribution. Further, the composition of classes in sampled patches should also be representative of spatial relationships between land cover types in the region of interest. **SEE** <https://chatgpt.com/c/6894f587-a918-832a-8112-73ab5926bb09>, <https://www.mdpi.com/2072-4292/15/6/1607#:~:text=Figure%205C%20we%20show%20that,the%20reported%20confusion%20matrix%20accuracies> This issue is exacerbated under low-data scenarios, where it can be difficult to capture the full diversity of both land cover types and their spatial relationships with few samples.

In this work, we demonstrate that employing a novel stratification strategy that groups candidate

patches according to their semantic content (i.e., land cover composition) using a coarse-resolution land cover dataset to guide the stratification selection such to sample patches that are representative of the diversity of land cover compositions found in the region of interest. Additionally, we employ Bootstrap Your Own Latent (BYOL) as a self-supervised pre-text task during pre-training, enabling a ResNet-101 convolutional encoder network to learn to extract deep features from aerial imagery without the need for full supervision using labelled data. This approach is used to tackle land cover classification at 1 m resolution over the state of Mississippi using only 1,000 patches selected using the semantic stratification procedure. We demonstrate that these approaches can be used to train deep semantic segmentation models that are accurate, generalizable, and capable of yielding acceptable performance even when supplied with only 250 patches of imagery-ground truth pairs during the supervised fine-tuning stage.

2 Related work

2.1 Approaches to very high spatial resolution land cover classification

maybe define VHSR in the intro???

VHSR land cover classification has historically been tricky due to blah blah blah Prior to deep learning, OBIA was the predominant paradigm - mention a few models Performance of OBIA at VHSR attributable to integration of spatial information into imagery classification Drawbacks, scale parameters, generalization, human intervention

Semantic segmentation introduced in computer vision community Why it worked well Prominence in VHSR classification today List a couple of notable works on convolution

2.2 Training data sampling strategies

Mention stratification, classical pixel-wise Struggles when applying to patch wise Mention a few

2.3 Remote sensing pre-training and self-supervised learning

Bring up pre-training, why its important, how transfer learning works Difficulty in remote sensing pre-training, unsure of domain shift between standard ImageNet pre-training and remote sensing imagery Emergence of self-supervised learning as a new paradigm for pre-training

How this has translated to remote sensing in the form of foundation models What these foundation models do, why they are built Problems with foundation models: ViT lacks inductive bias toward imagery Transformers still not the norm in remote sensing analysis, lack of inductive bias makes very low data fine-tuning difficult Still, can apply SSL to CNNs using contrastive/self-distillation approaches

3 Study area

The overarching goal of this proposed work is to develop a 1 m resolution land cover map for the state of Mississippi to facilitate high-fidelity land analysis. Mississippi is located in the southeastern

United States, and is home to a diverse range of land cover types. The region's humid subtropical climate and fertile soils in the Mississippi River Delta make these areas home to a variety of agricultural activities Mississippi State University Extension, while the remainder of the state is characterized by forests and woodlands which comprise the majority of land cover in the state United States Geological Survey. The coastal region along the Gulf of Mexico is home to a sprawling urban network intertwined with herbaceous wetlands and barren beaches English (2011). The state's diverse land cover types, geographical position along the Mississippi River and Gulf of Mexico, and economic dependence on agriculture and natural resources make the need for accurate land cover information at high spatial resolution a priority in order to support natural resource management, agricultural planning, and environmental monitoring efforts.

National Agriculture Imagery Program (NAIP) For this study, USDA National Agriculture Imagery Program (NAIP) imagery is used as the primary data source from which the models are trained and the land cover map is generated. NAIP imagery is high-resolution (0.6 meter) multispectral aerial imagery collected during the summer months over the contiguous United States. Such imagery is collected on a 2–3 year cycle (depending on the state) and is predominantly used for agricultural purposes. While the temporal availability of NAIP imagery is limited, the high spatial resolution of the imagery and public availability of the data make it an attractive data source for remote sensing applications that require high spatial resolution data over large areas in the contiguous United States Earth Resources Observation and Science (EROS) Center (2017). NAIP imagery collected in 2023 for the state of Mississippi is used in this study, which is the most recent data available at the time of writing. The imagery is provided in 4 spectral bands (Red, Green, Blue, and Near-Infrared) at a spatial resolution of 0.6 meters per pixel. The data is downloaded from the NRCS Geospatial Data Gateway, which provides color-infrared orthorectified, mosaiced imagery for each county for which NAIP imagery is available USDA et al. (2016). The imagery is resampled to a spatial resolution of 1 m to reduce the computational cost of processing and storing the data, as well as to align with the desired spatial resolution of the land cover product we aim to generate. Further, in accordance with recommendations from the Mississippi Automated Resource Information System (MARIS), the imagery is reprojected to the Mississippi Transverse Mercator (MSTM) coordinate system (EPSG: 3813) to ensure that all data is in the same coordinate reference system, as both UTM and State Plane coordinate systems divide the state of Mississippi into two zones, which can introduce complexity when working with datasets that span the entire state Mississippi Automated Resource Information System. Figure 1 shows the extent of the study area and the NAIP imagery used in this study in the MSTM coordinate system.

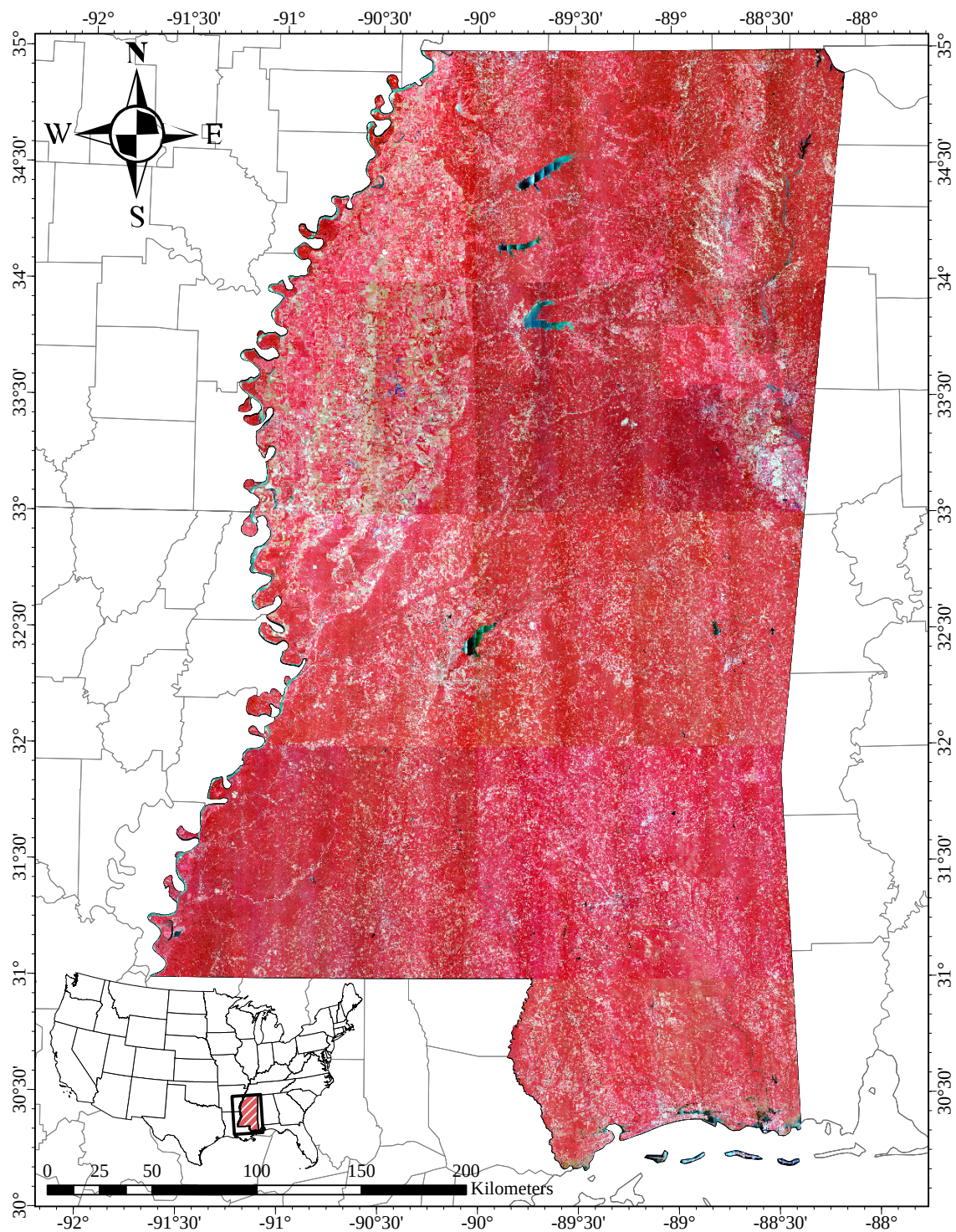


Figure 1: 2023 NAIP color-infrared imagery over Mississippi, USA.

4 Methodology

The proposed methodology consists of a multi-stage pipeline that leverages both self-supervised and supervised pre-training to develop a deep learning model for land cover classification over the state of Mississippi. Data for this proposed work consists of NAIP imagery over the state of Mississippi and Northern Virginia, as well as ground truth land cover data from the Chesapeake Bay Land Cover Dataset. For the Mississippi land cover tasks, a stratified sampling strategy is employed to select training, validation, and testing tiles for annotation that are representative of the study area using pre-existing coarse resolution land cover data to guide the stratification. The model architecture is based on U-Net, with a ResNet-152 encoder network and a decoder network that uses bilinear upsampling and residual connections to facilitate the flow of information through the network. The encoder of the model is pre-trained using a self-supervised denoising autoencoder approach, where a lightweight decoder network is used to reconstruct the input image from feature maps extracted by the encoder. The encoder network is then placed in a U-Net architecture and further pre-trained using supervised land cover data from Fairfax County, Virginia, before being fine-tuned on the Mississippi land cover data. Finally, the model is applied to the complete Mississippi NAIP dataset to generate a high-resolution land cover map for the state, and a thematic accuracy assessment is performed to evaluate the quality of the model’s predictions.

4.1 Training data generation

Mississippi High Resolution Land Cover For this study, a subset of the Mississippi NAIP imagery is utilized for the purposes of training and evaluating the deep learning models for land cover classification. Sampled tiles are annotated by a team of analysts using the Computer Vision Annotation Tool (CVAT) platform. CVAT is an open-source web-based tool that facilitates rapid, high-quality, dense annotations of imagery data using a variety of utilities, including the ability to use a pre-trained Segment Anything Model (SAM) 2 Ravi et al. (2024) to quickly generate initial segmentation masks for objects using point prompts that can be refined and assigned to specific classes by human annotators CVAT.ai Corporation (2023). The land cover legend used in this proposed methodology consist of 8 classes, with each class defined in Table 1. The inclusion of an “Unclassified” class is intended to capture areas of the image that are poorly illuminated or occluded by shadows or other objects, where the annotators are unable to confidently assign a class label. This is particularly important due to the presence of dense forests in Mississippi, which cause varying intensities of shadows to be cast on the ground, which makes classifying the underlying land cover impossible in some cases.

Table 1: Land cover classification scheme for the Mississippi dataset.

Index	Class	Description
1	Open Water	Rivers, lakes, ponds, and other standing or moving water. Water bodies typically appear dark in CIR imagery due to near-infrared absorption; may appear bluish-green depending on depth and turbidity.
2	Impervious Structures	Elevated built features (e.g., buildings, sheds). Bright and reflective with sharp-edges and shadows.
3	Impervious Surfaces	Paved areas such as roads and parking lots. Typically bright but may appear dark depending on material. Linear patterns and lack of shadows distinguish from barren land and impervious structures.
4	Barren Land	Exposed soil, sand, or rock with minimal vegetation. Bright or neutral in CIR imagery, often interleaved with herbaceous areas.
5	Forest/Woody Vegetation	Tree-covered areas with dense canopy. Deep red in CIR imagery due to high near-infrared reflectance. Uneven texture with large shadows.
6	Herbaceous/Low Vegetation	Short vegetation, including grasses and shrubs. Smoother texture than forest, lighter red in CIR imagery. Common in clearings and field edges.
7	Cultivated Crops	Actively farmed fields, often in distinct rows. Bright red or pink in CIR imagery at peak growth. Presence of rows or patterns distinguish from herbaceous areas.
8	Unclassified	Shadows or occlusions obscuring surface features. Deep black regions near trees and buildings.

National Land Cover Database (NLCD) Land cover data from the USGS National Land Cover Database (NLCD) is used to guide the stratification of the Mississippi NAIP imagery for the purposes of sampling training, validation, and test data. The NLCD is a multi-temporal land cover dataset produced by the Multi-Resolution Land Characteristics (MRLC) Consortium, which is a partnership between the USGS, NASA, and other federal agencies. The newest edition of the NLCD, Annual NLCD, provides annual land cover data at a 30 m spatial resolution for the entire United States, and is derived from Landsat satellite imagery using a combination of spectral and temporal information. Thus, while the NLCD provides a lower spatial resolution than the NAIP imagery used in this study, it is valuable for providing a broad overview of land cover types across the state of Mississippi, and can be used to guide the stratification of the NAIP imagery for the purposes of sampling training, validation, and test data. Prior to use in this study, level 1 NLCD data is reprojected to the MSTM coordinate system to match the NAIP imagery, and is reclassified to match the land cover classes used in the Mississippi dataset as shown in Table 2.

Table 2: Reclassification scheme for aligning NLCD land cover classes with our desired classification scheme.

NLCD Class Index	NLCD Class	Mississippi Class Index	Mississippi Class
11	Open Water	1	Open Water
23	Developed, Medium Intensity	2	Impervious Structures
24	Developed, High Intensity		
21	Developed, Open Space		
22	Developed, Low Intensity	3	Impervious Surfaces
31	Barren Land	4	Barren
41	Deciduous Forest	5	Forest
42	Evergreen Forest		
43	Mixed Forest		
90	Woody Wetlands		
52	Shrub/Scrub	6	Herbaceous
71	Grassland/Herbaceous		
81	Pasture/Hay		
95	Emergent Herbaceous Wetlands		
82	Cultivated Crops	7	Cultivated Crops
250	NoData	0	No Data

Chesapeake Bay Land Cover In addition to the NAIP imagery and the Mississippi land cover labels, auxiliary land cover data from the Chesapeake Bay Land Cover dataset is used during the supervised pre-training phase of the model. The 2022 edition of the dataset, which provides land cover information over the Chesapeake Bay watershed at 1 m ground sampling distance (GSD) for the 2017–2018 time period, is used in this study Chesapeake Bay Program (2023). Specifically, land cover data over Fairfax and Arlington counties and portions of Alexandria, Loudon, and Prince William counties in Northern Virginia are used to guide the supervised pre-training phase of the model. This region was chosen due to the availability high-resolution land cover data, as well as the presence of dense urban and suburban land cover types that are not well-represented in the Mississippi NAIP dataset. Corresponding 2018 color-infrared NAIP imagery for this region is also used in this study, and is provided by the NRCS Geospatial Data Gateway. It is important to note that the orthoimages provided by the NRCS are not precisely clipped to the boundaries of the counties, and are instead provided as a single, large mosaic that includes multiple images with some overlap between adjacent counties. The land cover data is reprojected to the UTM Zone 18N coordinate system (EPSG:32618) to match the imagery, and the NAIP imagery is subsequently resampled to 1 m GSD such to align with the spatial resolution of the land cover data. Further, the land cover data is reclassified to more closely match the land cover classes used in the Mississippi dataset, as shown in Table 3. The full extent of the subset of the Chesapeake Bay dataset used in this study is shown in Figure 2. In total, 10,220 tiles are used for the supervised pre-training phase of the model, while 5,110 tiles are used for validation, and 5,109 tiles are used for testing.

Table 3: Reclassification scheme for reducing Chesapeake Bay land cover classes to more closely align with our desired classification scheme.

Chesapeake Class Index	Chesapeake Class	Reduced Class Index	Reduced Class Label
1	Water	1	Open Water
3	Tree Canopy	2	Forest
10	Tree Canopy over Impervious Structures		
11	Tree Canopy over Other Impervious		
12	Tree Canopy over Roads		
4	Shrubland	3	Herbaceous
2	Emergent Wetlands	4	Herbaceous
5	Low Vegetation		
6	Barren Land	5	Barren
7	Impervious Structures	6	Impervious Structures
8	Other Impervious	7	Impervious Surfaces
9	Roads		

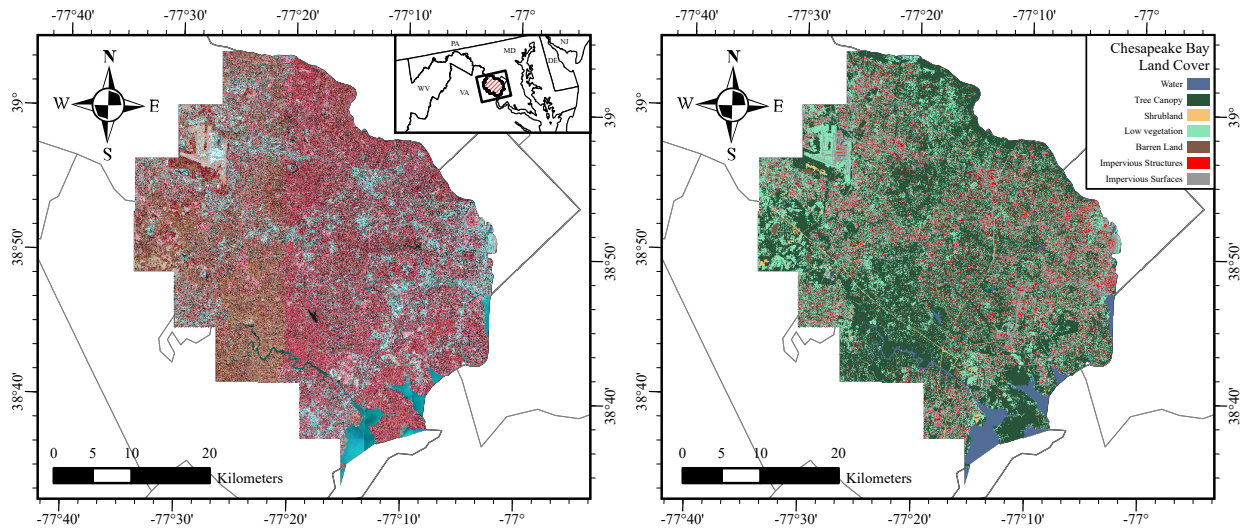


Figure 2: Extent of 2018 NAIP imagery (left) and 2018 Chesapeake Bay land cover (right) over Alexandria, Arlington, Fairfax, Loudon, and Prince William counties in Virginia, USA used in this study.

4.2 Stratifying by land cover composition

Ideally, land cover classification models should be trained and validated against samples that are representative of the study area. However, under the constraints of limited labeled data, it may be difficult to capture the full range of variation in the study area. Stratified sampling is a common approach to address this issue, where the study area is divided into strata based on some characteristic (such as land cover class or sub-region) and samples are drawn from each stratum. Unlike traditional land cover classification models, semantic segmentation models necessitate the use of densely annotated tiles that exhibit a high degree of diversity with regards to spatial characteristics. Ideally, a single training tile should consist of a mix of land cover classes, and the arrangement of land cover classes should similarly be representative of the study area as a whole.

To achieve this, a stratified sampling strategy is employed using the NLCD to guide the selection of training, validation, and testing tiles. This ensures that both the distribution of classes in the tiles is representative of the study area and the content of the tiles is heterogeneous and diverse in terms of spatial characteristics.

A grid of 256×256 m square polygons is overlaid over the study area to create candidate tiles, and the proportion of each land cover class is calculated for each tile by computing zonal statistics on the NLCD data within each tile. Once the relative frequency of each land cover class is calculated, these frequencies are scaled by subtracting the median value and dividing by the interquartile range of each class. Principal component analysis (PCA) is applied to the scaled land cover relative frequency data to reduce the dimensionality of the relative prior to clustering. Once the principal components are calculated, the first two principal components are identified as capturing 94.24% of the data variance; subsequently, these principal components are used to guide the stratification of the data. Figure 4 visualizes these first two principal components of the land cover data at the tile level. Higher values of the first principal component correspond to a higher proportion of impervious land cover types in each tile, while higher values of the second principal component correspond to a higher proportion of forested and herbaceous land cover, with lower values correlating with regions that consist of tiles primarily or wholly composed of cultivated crops. These tiles are subsequently grouped into 250 clusters using K-means clustering operating on these two principal components. As shown in Figure 5, the clustering algorithm effectively divides the principal component space into regions that correspond to differing land cover compositions. From each cluster, 2 candidate tiles are sampled for training, 1 for validation, and 1 for testing. This results in a total of 500 training tiles, 250 validation tiles, and 250 testing tiles. From the remaining candidate patches, 20% of the tiles are randomly sampled for pre-training and 5% are sampled for pre-training validation, yielding 377,913 and 94,480 tiles, respectively.

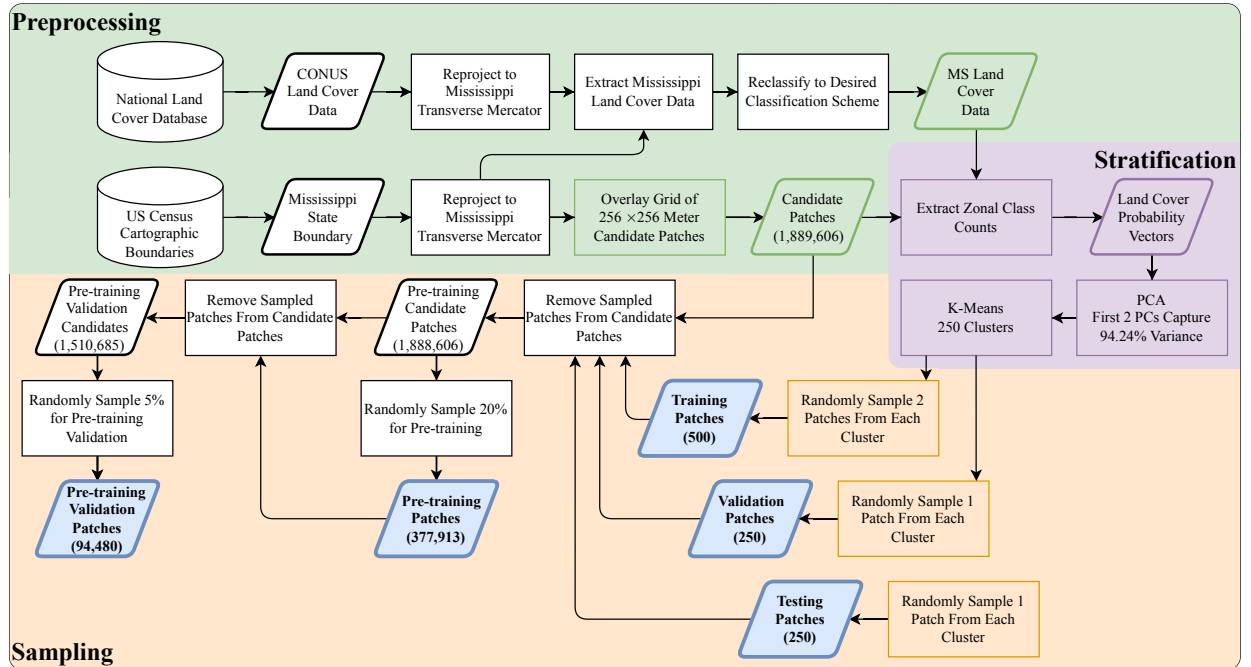


Figure 3: Flowchart of the stratified sampling process for selecting data for model training and validation.

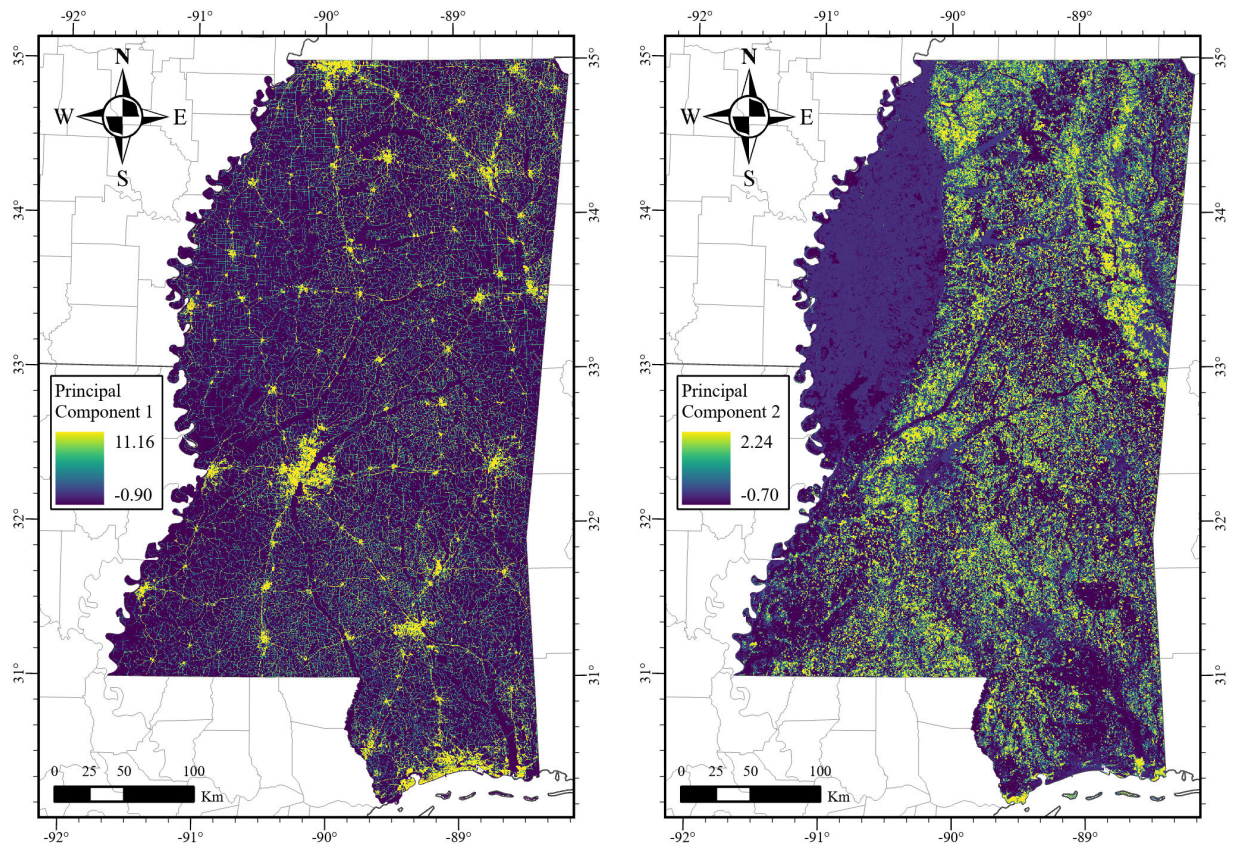


Figure 4: Principal components of NLCD relative frequencies per candidate tile.

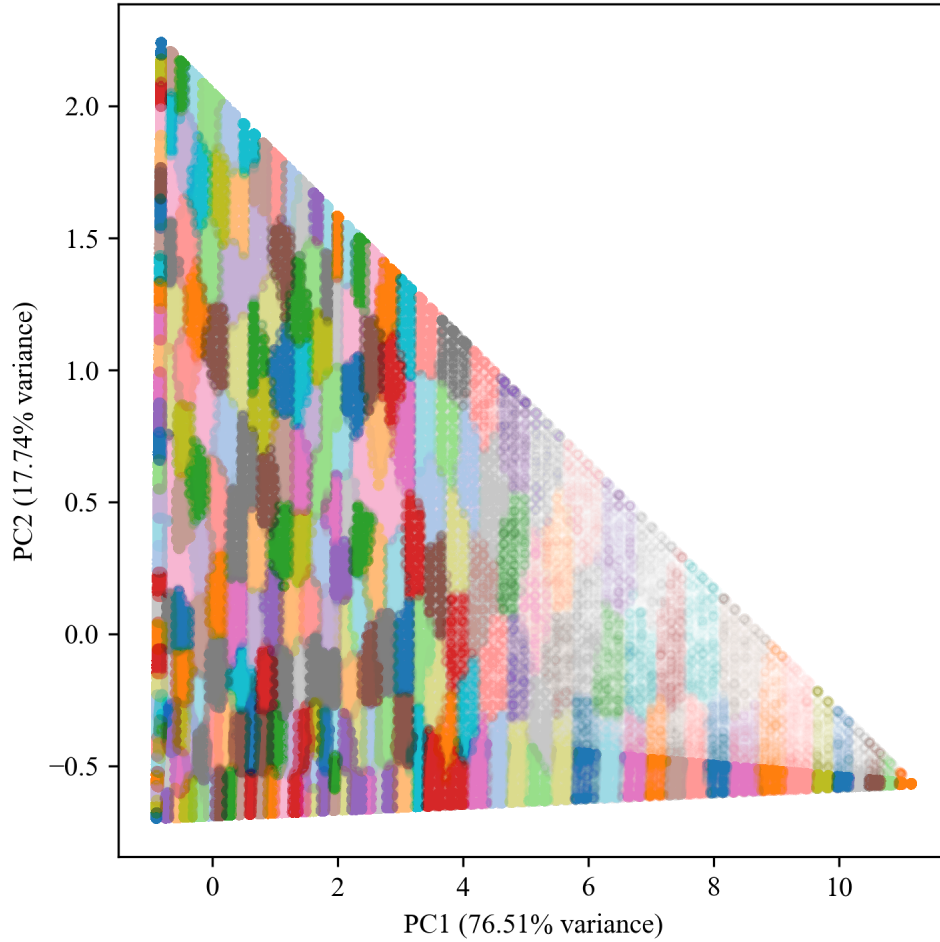


Figure 5: Clusters from which samples are draw in principal component space.

Ensuring this strategy yields high-quality training data that is representative of the study area is crucial to the success of the model. However, a rigid evaluation of the stratification approach is not possible due to the lack of ground truth data for the study area. One measure that can be taken to ensure the stratification is effective is to inspect the distribution of land cover classes across the splits. Figure 6 shows that the distribution of land cover classes is consistent across the training, validation, and testing splits, which is ideal for ensuring that the model is exposed to a representative sample of the study area’s land cover diversity. One important note is that the distribution of land cover classes in the study area is highly imbalanced, with the majority of Mississippi’s land cover being classified as forest, and a low density of urban and developed areas. To tackle this issue, other strategies often employ synthetic oversampling of minority classes to force the distribution of classes into a more balanced state Douzas et al. (2019); Fonseca et al. (2021). Although the proposed approach to stratifying the data does not explicitly enforce such policies, comparing the land cover class distribution in the sampled tiles to the distribution across the entire state (as shown in Figure 6) demonstrates that this sampling strategy achieves a similar outcome to such strategies. Furthermore, this approach tends to avoid sampling tiles that are composed of a single land cover class. In the original NLCD data, 29.96% of the tiles are composed of a single land cover class, while only 1.90% of the sampled tiles are composed of a single land cover class. This is promising,

as heterogeneous tiles with a mixture of land cover classes convey information about the spatial organization of land cover classes. Additionally, a visual inspection of the location of the sampled tiles in Figure 7 shows that the tiles are distributed across the state of Mississippi without any apparent spatial bias.

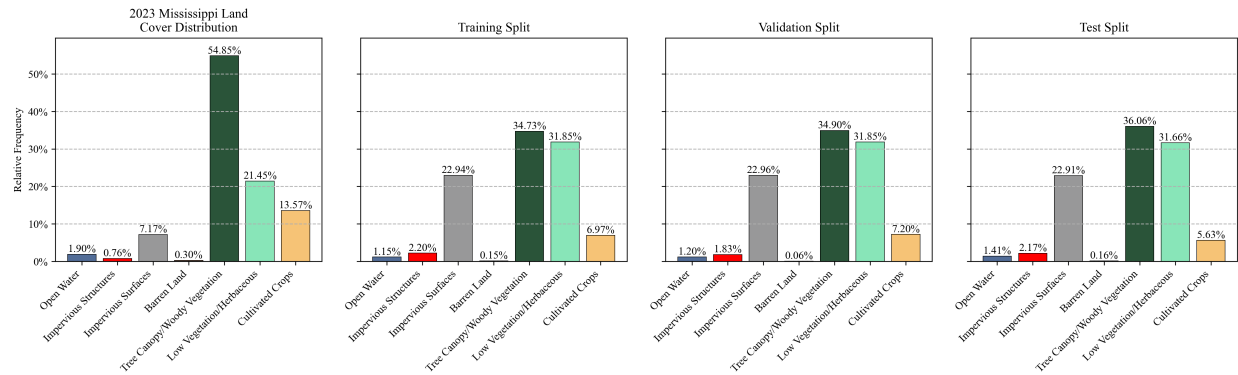


Figure 6: Distribution of 2023 NLCD land cover classes across Mississippi: statewide distribution compared with training, validation, and test splits.

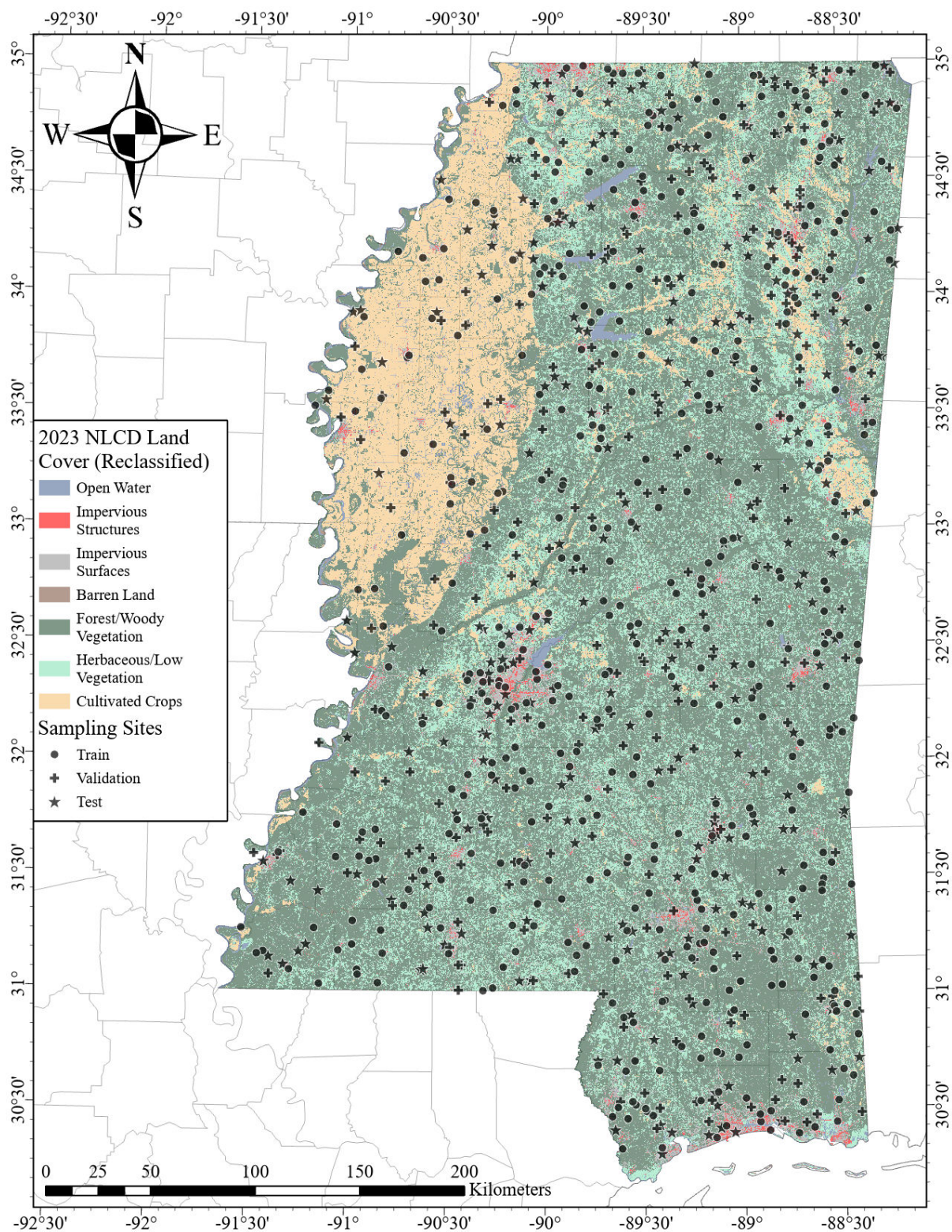


Figure 7: Sample locations for the Mississippi dataset.

4.3 Residual U-Net model configuration

The proposed model is based on the U-Net architecture, which is a popular choice for semantic segmentation tasks due to its relatively simple yet effective design and its popularity in the remote sensing community Li et al. (2018); Pan et al. (2020b); Robinson et al. (2019); Ronneberger et al. (2015). Several modifications are made to the original U-Net architecture to improve the model’s performance. First, in lieu of the original convolutional encoder network in the original implementation, a much deeper ResNet-152 network is used in its place. The reasoning behind this choice is that the increased number of layers in the ResNet-152 network facilitate more complex feature extraction, particularly during the self-supervised pre-training phase, where the large amount of unlabeled data will allow the encoder to learn to capture rich representations of spatial features in the input imagery. The number of channels accepted and produced by the decoder convolutions are subsequently adjusted to match the output of their corresponding encoder blocks. Second, the transposed convolutional layers in the original U-Net decoder are replaced with bilinear upsampling to reduce the presence of checkerboard artifacts in the upsampled feature maps, which can inhibit the quality of the output segmentation maps Odena et al. (2016); Sanjar et al. (2020). Finally, in each decoder block, residual connections are used to facilitate the flow of information through the network and improve training stability He et al. (2015).

4.4 Multi-stage pre-training approach

The proposed pre-training approach consists of two phases: a self-supervised pre-training phase and a fully supervised pre-training phase. As shown in Figure 8, the model is first trained using the unannotated NAIP imagery over the state of Mississippi, then additional pre-training is performed using NAIP imagery over Fairfax County, Virginia using ground truth land cover data from the Chesapeake Bay Land Cover Dataset, before being fine-tuned on our hand-labeled training data from Mississippi. The purpose of such a pre-training regimen is to leverage the large volume of imagery available to train the encoder network to extract features that are useful for image reconstruction as well as relevant for land cover classification Vincent et al. (2008). The second phase of pre-training encourages the model to refine these features in a supervised setting, while also initializing the decoder network using a land cover classification task that is nearly aligned with the downstream Mississippi land cover classification task Martins et al. (2022); Wurm et al. (2019). The goal of such an approach is to reduce the amount of labeled data required to achieve optimal performance on the final land cover classification task Hu et al. (2015); Lagrange et al. (2015); Weiss et al. (2016); Yosinski et al. (2014).

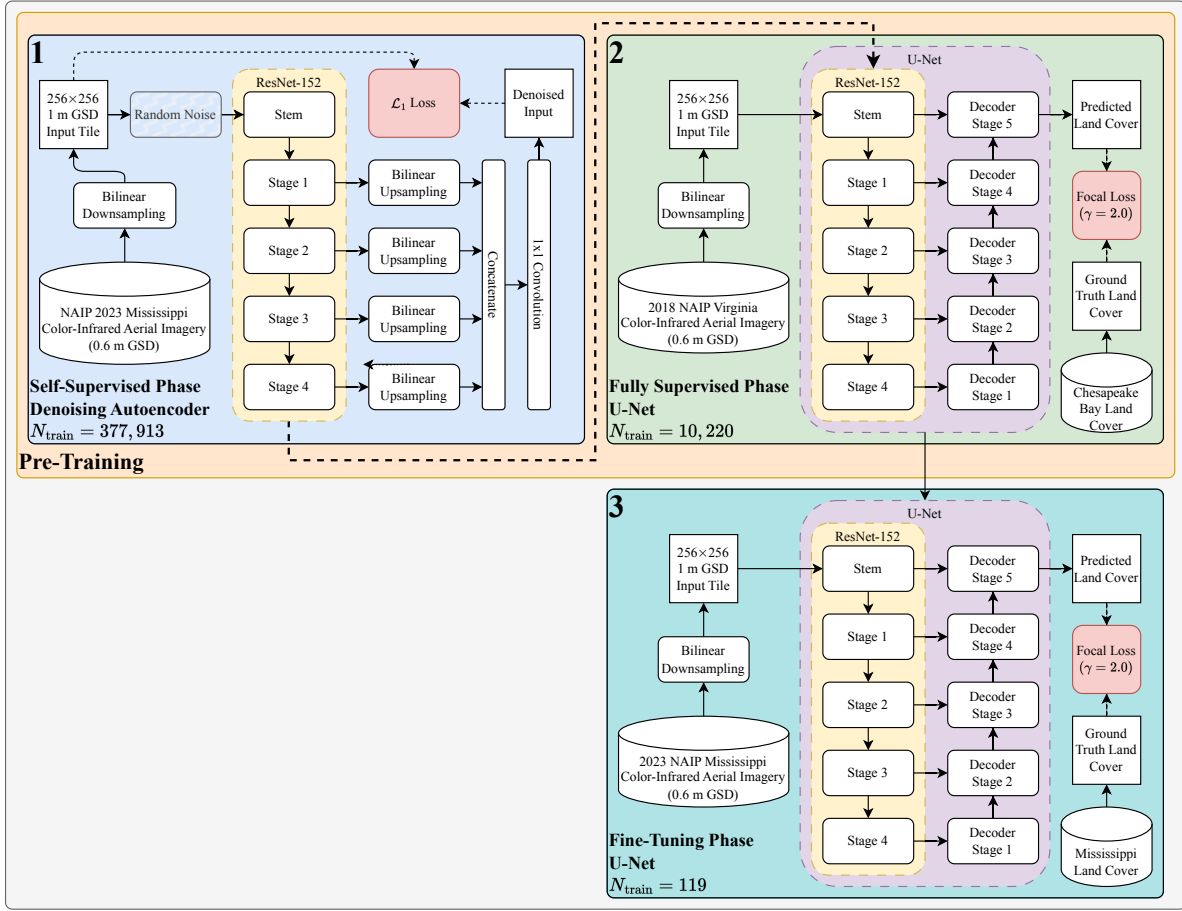


Figure 8: Flowchart of the model training workflow.

Self-supervised pre-training In the first stage of pre-training, a denoising autoencoder is trained using the unannotated NAIP imagery over the state of Mississippi. This denoising autoencoder consists of a ResNet-152 encoder network and a simple decoder network that receives feature maps from each stage of the encoder network, resamples them to the original input resolution using bilinear interpolation, concatenates them, and passes them through a single 1×1 convolutional layer to produce an output with the same dimensions as the input. The ResNet-152 encoder network is initialized with ImageNet-1k pre-trained weights, while the decoder network is randomly initialized. A full decoder network (as is common in traditional autoencoders) is not used in this phase, as the primary goal of this stage of pre-training is to force the encoder network to learn robust features present in the imagery that are necessary for reconstruction, rather than to necessarily learn to denoise the input images with a high degree of accuracy. The single-layer decoder network is designed to ensure that backpropagation primarily influences the parameters of the encoder network, while also reducing the computational cost of pre-training the model.

Two types of noise are added to the image: Gaussian noise and Poisson noise. The intensity of the Gaussian noise is controlled by the standard deviation parameter σ , which is randomly sampled from a uniform distribution $U(0, 1)$. Similarly, the intensity of the Poisson noise is controlled by the rate parameter λ , which is also randomly sampled from a uniform distribution $U(0, 1)$. The

model inputs are standardized to have zero mean and unit variance on a per-band basis using the mean and standard deviation of the pre-training data prior to adding noise. The model is trained using Layer-wise Adaptive Rate Scaling (LARS) optimizer in order to stabilize training with a large batch size of 1024 You et al. (2017). After each epoch of pre-training, the model is evaluated using the pre-training validation set to ensure that the model is fitting the data properly and to adjust the learning rate if necessary. If the validation loss does not decrease after 5 epochs, the learning rate is reduced by a factor of 0.1. If the validation loss does not decrease after 15 epochs, the training is terminated early.

Fully supervised pre-training Once the self-supervised pre-training phase is complete, the encoder network is used as the encoder in the U-Net architecture, and the decoder network is initialized with random weights. The model is trained to minimize the Focal loss, defined as:

$$\mathcal{L}_{\text{Focal}}(p_c) = \begin{cases} -\alpha_c (1 - p_c)^\gamma \log(p_c) & \text{if } y = c \\ -(1 - \alpha_c) p_c^\gamma \log(1 - p_c) & \text{otherwise} \end{cases} \quad (1)$$

where p_c is the predicted probability of class c , y is the true class label, α_c is the class weight for class c , and γ is a hyperparameter that controls the degree of class weighting. In order to address the class imbalance present in the data, class weights are assigned according to the following formula:

$$\alpha_c = \frac{C (1 - \rho_c)^\beta}{\sum_{c=1}^C (1 - \rho_c)^\beta} \quad (2)$$

where ρ_c is the proportion of class c in the training set, β is a hyperparameter that controls the degree of class weighting, and C is the total number of classes. γ is set to 2.0 in accordance with the authors' findings in the original Focal loss implementation Lin et al. (2017), and $\beta = 2.0$ such that minority classes are emphasized without majority classes being under-represented in the loss function. The model's parameters are optimized using the AdamW optimizer with a batch size of 64 and an initial learning rate of 1e-4. The model is evaluated after every epoch using the validation set to adjust the learning rate and terminate the training early if necessary: if no improvement is observed in the validation loss for 3 consecutive epochs, the learning rate is reduced by a factor of 0.1, and if no improvement is observed for 15 consecutive epochs, the training is terminated early. During this phase, the model is trained using two different configurations: one where the entire model is trained end-to-end, and one where only the decoder network is trained while the encoder network is frozen. Three different sets of encoder parameters are also experimented with: ImageNet-1k pre-trained weights, randomly initialized weights, and weights pre-trained using the self-supervised pre-training phase. After each model has been trained, they are evaluated using the withheld testing set to examine the performance of each training configuration on the Fairfax County land cover classification task. Evaluating the model on this task, while not directly related to the downstream Mississippi land cover classification task, gives an indication of the model's ability to learn relevant features from the data and to generalize to a new, but similar, land cover classification task.

Fine-tuning During the final fine-tuning phase, the model is trained using the labeled data from the state of Mississippi. The same loss function, class weighting strategy, optimizer, and

hyperparameters as the fully supervised pre-training phase are employed. Several sets of initial weights are tested, including random initialization, ImageNet-1k pre-trained weights in the encoder network, and weights pre-trained using the self-supervised and/or fully supervised pre-training phases. Further, for each model configuration, three different scenarios are tested: one where the entire model is trained end-to-end, one where the encoder parameters are frozen and only the decoder is trained, and one where the entire model is frozen and only the final classification layer is trained. After training, the model is evaluated using the holdout testing set to examine the model’s efficacy on the Mississippi land cover classification task.

4.5 Wide area inference for large-scale land cover mapping

Once the best performing model has been selected based on the testing set performance (specifically the macro F1 score), the model is integrated into an inference pipeline to generate land cover classification maps for the entire state of Mississippi. As the NAIP imagery is split into county-level subsets (see Section 3), each county is processed individually, then the resulting land cover classification maps are stitched together to create a single, seamless land cover classification map for the entire state. Prior to performing inference on a county, the county’s boundary is obtained from the US Census Bureau’s TIGER/Line Shapefiles, buffered by 1.5 times the model input size to ensure sufficient context for accurate predictions near the county boundary, and the NAIP imagery is cropped to the buffered boundary. A “sliding window” approach is applied to inference, where the model processes overlapping patches of the cropped NAIP imagery, and the resulting probabilities are post-processed and averaged to create a single land cover classification map for the county. For the inference pipeline, a model input size of 256×256 pixels and a stride of 128 pixels are selected, which allows for a 50% overlap between adjacent patches. To reduce the impact of edge effects on the stitched land cover classification map, the probabilities of each pixel are weighted by their distance from the center of the tile by multiplying the probabilities by a 2D Gaussian kernel with the same dimensions as the model input size with $\sigma = 128$. This ensures that the probabilities of pixels near the center of the tile have a greater impact on the final land cover classification map than those near the edges of the tile. A weighted average of the probabilities of each pixel with the same coordinates in each tile is performed to create the final land cover classification map for the county.

4.6 Model evaluation and accuracy assessment

To evaluate the performance of the proposed model, several metrics will be utilized: overall accuracy, producer’s accuracy, user’s accuracy, F1 score, and Cohen’s Kappa. Due to the severe class imbalance present in the dataset, the model performance will be measured on both a per-class basis (using macro averaging) and a per-pixel basis. The model exhibiting the highest macro-averaged F1 score on the testing set will be selected to generate the land cover classification map for the entire state of Mississippi. A thematic accuracy assessment of the resulting land cover classification map will be conducted using a random sampling approach, selecting 10,000 points across Mississippi. Regions previously used as part of the pre-training, fine-tuning, validation, or testing sets will be excluded from this accuracy assessment to ensure evaluation on unseen data. Additionally, to reduce the impact of spatial autocorrelation on the accuracy assessment, a minimum distance of 256 m will be enforced between sampled points.

5 Preliminary Results and Discussion

To collect preliminary results, 250 tiles from the full Mississippi land cover dataset were used for fine-tuning training, validation, and testing in the last stage of the multi-stage pre-training approach. In total, 119 tiles were used for training, 61 tiles were used for validation, and 70 tiles were used for testing. These early results give insight into the efficacy of this approach: Table 4 shows that introducing a denoising autoencoder pre-training stage prior to supervised pre-training has a substantial impact on a model’s performance. This is evidence that the core idea of the proposed approach has merit, in that the denoising autoencoder pre-training stage enables the model to learn spatial features that are useful for land cover classification, even when the model undergoes subsequent supervised pre-training on a different region. Additionally, the choice of which layers to freeze during the fully supervised pre-training stage has a substantial impact on the model’s performance. As shown in Table 5, the most effective strategy for freezing parameters is to train the full network during the fully-supervised pre-training stage, but freeze the encoder network during the final fine-tuning stage. Examining the results of the Chesapeake Bay Land Cover classification task (i.e., the second stage of the multi-stage pre-training approach) in Table 6 shows more meager results compared to the Mississippi Land Cover classification task, with the denoising autoencoder pre-training stage providing only a marginal improvement over ImageNet pre-training. This suggests that the denoising autoencoder pre-training stage is more effective when the model undergoes further pre-training in a supervised manner. Further, this opens the door to future work on using a stronger pre-training strategy for the first stage of the multi-stage pre-training approach, such as increasing the noise level in the denoising autoencoder or using a more complex pre-text task to further improve the model’s performance.

Table 4: Performance comparison of land cover classification models on Mississippi test data showing how different pre-training strategies (random initialization, ImageNet weights, and ImageNet weights followed by denoising autoencoder pre-training) affect accuracy metrics when all models undergo the same two-stage fine-tuning process: full encoder training on Chesapeake Bay data followed by decoder-only fine-tuning on Mississippi data with frozen encoders.

Stage 1 Pre-Training Strategy	Overall Accuracy (%)	Overall F1 Score (%)	Macro F1 Score (%)	Loss
Random Initialization	28.96	30.47	10.34	36044
ImageNet	77.54	77.58	50.25	19638
ImageNet + Denoising Autoencoder	82.11	82.62	70.45	11364

Table 5: Effects of training on different portions of the network during the fully supervised training stages.

Weights Trained					Loss
Fully Supervised Pre-training Encoder	Decoder	Encoder	Decoder	Classifier Head	
	✓			✓	38126
	✓		✓	✓	19725
	✓	✓	✓	✓	14511
✓	✓			✓	38949
✓	✓		✓	✓	11364
✓	✓	✓	✓	✓	11725

Table 6: Performance metrics for models trained on the Chesapeake Bay Land Cover dataset with different pre-training strategies.

Pre-training strategy	Overall Accuracy (%)	Macro F1 Score (%)	Loss
None	77.51	52.59	11056
ImageNet	82.50	67.16	7393
ImageNet + Denoising Autoencoder	82.63	67.43	7439

The best model from the multi-stage pre-training approach was used to classify land cover in the state of Mississippi at 1 m resolution in order to understand the strengths and limitations of the proposed approach. A visual assessment of the land cover classification results for the Mississippi dataset is shown in Figure 9. The confusion matrix for the land cover classification results is shown in Table 5, and the classification report is shown in Table 8. These results show that, overall, the model is fairly accurate, even with the limited labeled data available for training. However, more data is needed to improve the model’s performance on certain classes, such as cultivated crops, barren land, and open water. This suggests that, despite the promising results of the multi-stage pre-training approach, more labeled data is needed to fully leverage the feature learning capabilities of the model and improve its performance on classes that are underrepresented in the training data.

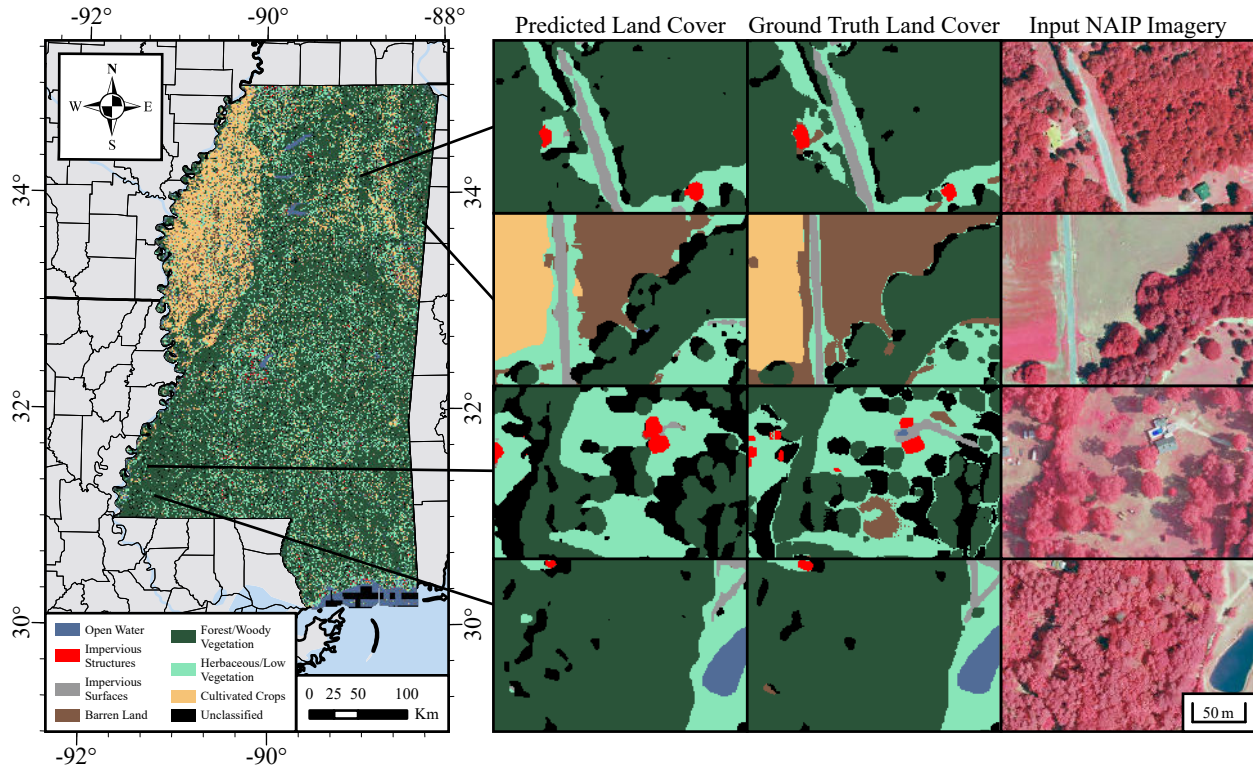


Figure 9: High-resolution land cover classification results for the Mississippi dataset (left) alongside the ground truth labels (middle) and the false-color NAIP imagery used as input (right).

Table 7: Confusion matrix for the Mississippi land cover classification results.

Predicted Label \ True Label	Open Water	Impervious Structures	Impervious Surfaces	Barren Land	Tree Canopy/ Woody Vegetation	Low Vegetation/ Herbaceous	Cultivated Crops	Unclassified	Producer's Accuracy
Open Water	7154	344	41	45	16	141	0	1066	81.23%
Impervious Structures	13	20776	2041	1335	115	4891	0	1708	67.28%
Impervious Surfaces	19	2590	124159	9058	1722	14407	13	3225	80.00%
Barren Land	298	1964	21482	130287	2798	100670	24151	4008	45.61%
Tree Canopy/ Woody Vegetation	924	244	1403	1064	2336744	94750	26166	101005	91.20%
Low Vegetation/ Herbaceous	469	2596	24339	57498	58379	851272	136447	29850	73.33%
Cultivated Crops	0	0	171	3079	6685	48908	148924	603	54.73%
Unclassified	208	940	3546	384	14128	8236	106	147915	62.64%
User's Accuracy	78.75%	70.54%	70.07%	64.26%	96.53%	75.78%	44.34%	51.11%	Overall Accuracy 82.11%

Table 8: Per-class F1 scores and metrics for Mississippi land cover classification results.

Class	F1 Score (%)	Support
Open Water	79.97	8,807
Impervious (raised structures)	68.87	30,879
Impervious (paved surfaces)	74.71	155,193
Barren Land	53.35	285,658
Tree Canopy/Woody Vegetation	93.79	2,562,300
Low Vegetation/Herbaceous	74.54	1,160,850
Cultivated Crops	54.73	208,370
Unclassified	63.64	175,463
Total Support		4,587,520
Macro Metrics		
User's Accuracy (%)		68.93
Producer's Accuracy (%)		74.30
F1 Score (%)		70.45
Overall Metrics		
Accuracy (%)		82.12
User's Accuracy (%)		84.06
Producer's Accuracy (%)		82.12
F1 Score (%)		82.53
Cohen's κ		0.711

In conclusion, this research demonstrates that a multi-stage pre-training approach significantly enhances land cover classification performance under label-scarce conditions. Early results show that the proposed approach is an effective strategy for producing high spatial resolution land cover maps with limited labeled data. Beyond the immediate applications of the high spatial resolution land cover map for the state of Mississippi, this research has broader implications for land cover classification due to the unique approach to pre-training that opens the door for the development of more label efficient deep learning models. The proposed approach has the potential to enable high spatial resolution land cover classification at scale while minimizing the need for human annotation to produce ground truth data, paving the way for high resolution analyses quickly and efficiently. This is particularly important for situations where high spatial resolution land cover data for a region is necessary but existing datasets are scarce, outdated, or non-existent.

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